

ENGINEERING FOUNDATIONS • CLOUD
TRACK

Google Cloud Platform Fundamentals

Infrastructure, Projects,
Regions, and IAM for Beginners

~2h 30m / Lecture by: Imran Sayyed

GCP HIERARCHY

Organization



Projects



Resources

WHY IT MATTERS

Why GCP?

Same infrastructure

Runs on the network that powers Search, Gmail, YouTube, and Maps.

Global reach

Data centers and a private fiber network span every continent.

Pay as you go

Rent compute and storage by the second instead of buying servers.

Search

•

Gmail

•

YouTube

•

Maps

→

Google Cloud

THE BASICS

Introduction to GCP

Google Cloud Platform is a suite of on-demand computing services delivered over the internet.

Compute — virtual machines and functions

Storage — files, objects, and databases

Analytics — process data at huge scale

YOU PROVIDE

Your Code & Data



GOOGLE PROVIDES

Servers & Networking

Data Centers

Maintenance & Security

Major cloud providers

PROVIDER	AWS	Azure	GCP
Owner	Amazon	Microsoft	Google
Compute	EC2	Virtual Machines	Compute Engine
Object storage	S3	Blob Storage	Cloud Storage
Known for	Breadth	Enterprise	Data & AI

THE PHYSICAL LAYER

Google global infrastructure

Data centers

Huge buildings full of servers, power, and cooling that run your workloads.

Undersea cables

Fiber-optic cables across oceans link continents at light speed.

Regions

Groups of data centers placed close to users around the world.

Servers



Data Center



Region



Global Network

What is a region?

A region is a specific geographic location where Google operates data centers.

You choose a region close to your users to reduce delay and meet data residency rules.

EXAMPLE REGIONS

us-central1

Iowa

europa-west1

Belgium

asia-south1

Mumbai

asia-east1

Taiwan

What is a zone?

A zone is an isolated location inside a region — usually a single data center building.

Each region contains multiple zones so a failure in one zone does not take down the others.

Region: us-central1

Zone a

Zone b

Zone c

One region, three independent zones.

Region vs zone

Region

- A geographic area, e.g. Mumbai
- Contains several zones
- Picks how close you are to users

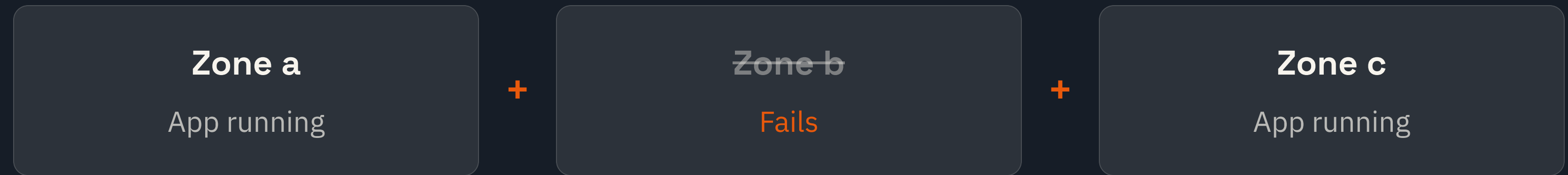
Zone

- One isolated location in a region
- Roughly a single data center
- Where a VM actually runs

1 Region ➤ Many Zones

Why multiple zones exist

Spreading copies of your app across zones keeps it online when one zone fails.



Result: users stay served — high availability

HOW RESOURCES ARE ORGANIZED

GCP hierarchy overview

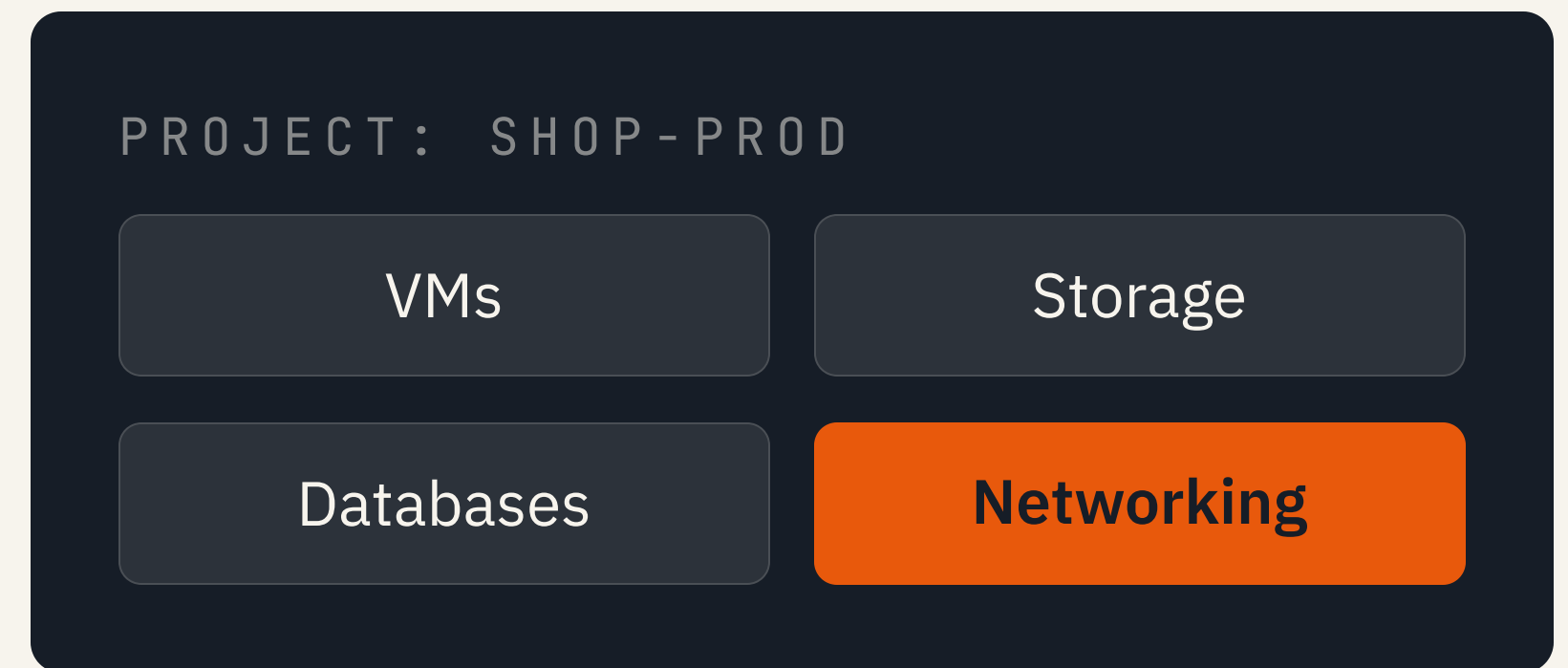


Policies set at the top flow down to everything beneath.

What is a project?

A project is the basic container that holds and organizes all your GCP resources.

Every resource lives in exactly one project, and billing and access are tracked per project.



WHY IT MATTERS

Why projects matter

Isolation

Separate dev, test, and prod so they never interfere with each other.

Billing

See exactly what each team or product costs, broken down per project.

Access

Grant people permissions on one project without exposing the rest.

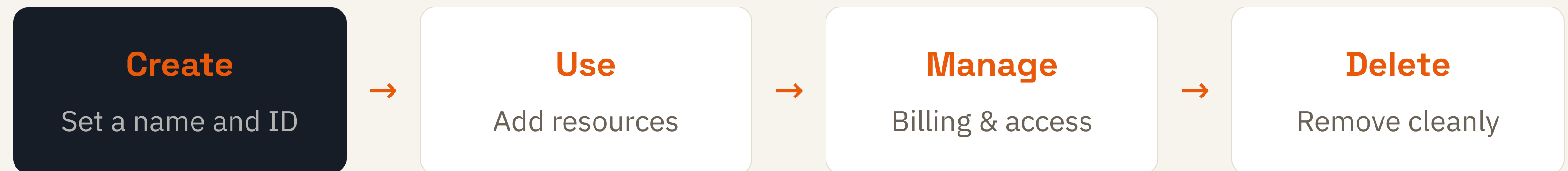
project-dev

project-test

project-prod

FROM START TO FINISH

Project lifecycle

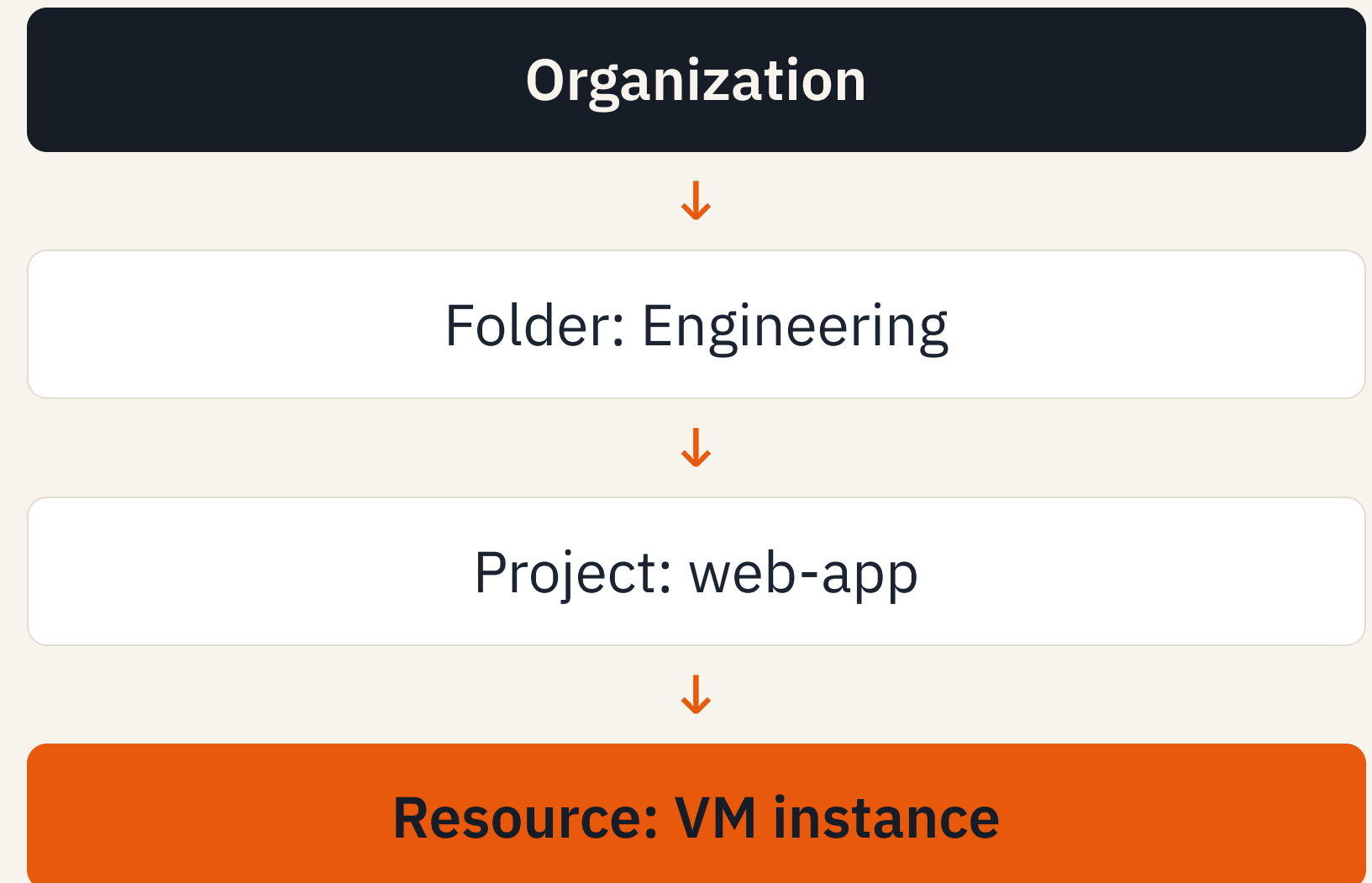


Deleting a project removes every resource inside it.

Resource management

A resource is anything you create in GCP — a VM, a storage bucket, or a database.

Resources are grouped by project, and projects are grouped by folders and organization.



Billing concepts

Billing account

Holds your payment method and pays the bills.



Projects

One billing account can fund many projects.

Pay only for what you use

Set budgets and alerts

No upfront hardware cost

STAYING ORGANIZED

Labels and tags

Labels are key-value pairs you attach to resources to sort and filter them.

They make it easy to group costs by team, environment, or product.

LABELS

env = production

team = payments

cost-center = web-2026

Introduction to IAM

IAM — Identity and Access Management — controls who can do what on which resource.

It is how GCP decides whether to allow or deny every single action.

The IAM Question

WHO

can do WHAT

on WHICH resource

WHY IT MATTERS

Why IAM matters

Security

Stops the wrong people from touching sensitive systems and data.

Accountability

Every action is tied to an identity, so you know who did what.

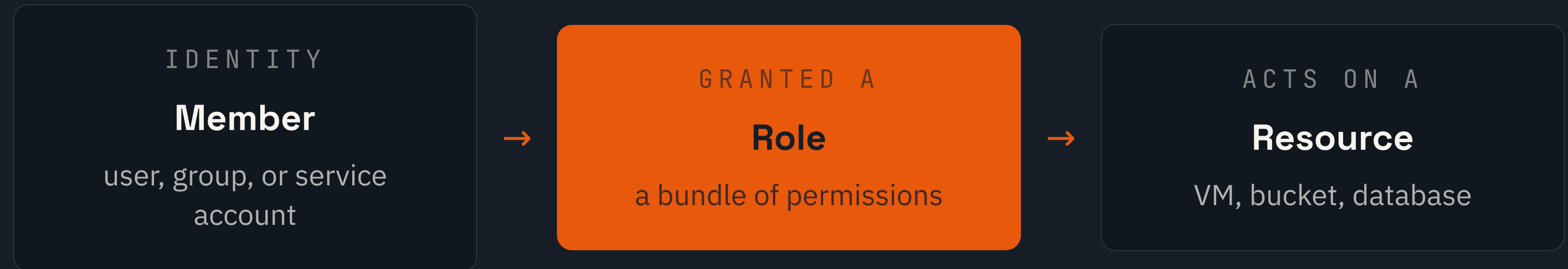
Control

Give each person exactly the access they need — nothing more.

Misconfigured access is one of the most common causes of cloud breaches.

HOW IAM WORKS

IAM architecture



Member + Role + Resource = Policy

THE BUILDING BLOCKS

IAM components

Member

The identity asking to act — a person, group, or service account.

Permission

A single allowed action, like reading a file or starting a VM.

Role

A named bundle of permissions you grant all at once.

Policy

The binding that says which member has which role on a resource.

THE STARTER ROLES

Basic roles

Owner

Full control — manage resources, billing, and access.

Editor

Can create and change resources, but not manage access.

Viewer

Read-only — can look at resources but change nothing.

Viewer< Editor< Owner

THE GOLDEN RULE

Principle of least privilege

Give every identity the smallest set of permissions it needs to do its job — and no more.

Prefer narrow predefined roles over the broad Owner or Editor roles.

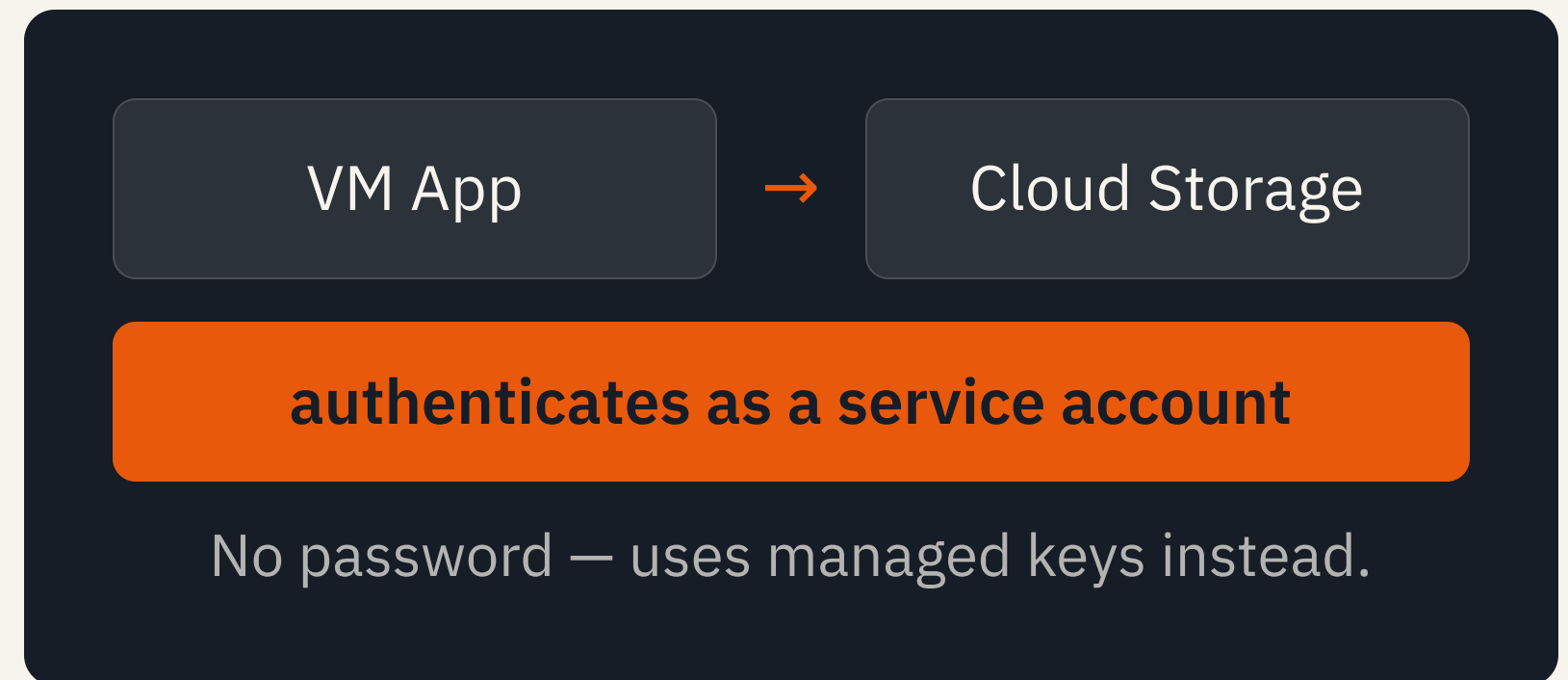


Grant from the bottom up, not the top down.

Service accounts

A service account is a special identity used by applications and VMs, not people.

It lets one service securely call another without a human logging in.



User vs service accounts

User account

- Belongs to a human
- Logs in with a password
- Used in the console

Service account

- Belongs to an app or VM
- Authenticates with keys
- Used in automation

YOUR CONTROL PANEL

The GCP Console

Web Console

A browser dashboard to click through and manage every service.

gcloud CLI

A command-line tool to script and automate the same actions.

APIs & SDKs

Call GCP directly from your code in any major language.

Three doors into the same platform – pick what fits the task.

FINDING YOUR WAY

Console dashboard

☰ Menu

Project: shop-prod

Search

Navigation menu

Jump to any service

Project selector

Switch between projects

Billing & IAM

Costs and access in one place

Compute Engine

Compute Engine gives you virtual machines — full computers you rent and run in the cloud.

- Pick CPU, memory, and disk size
- Choose a region and zone
- Pay per second while it runs

VM INSTANCE

Your Application

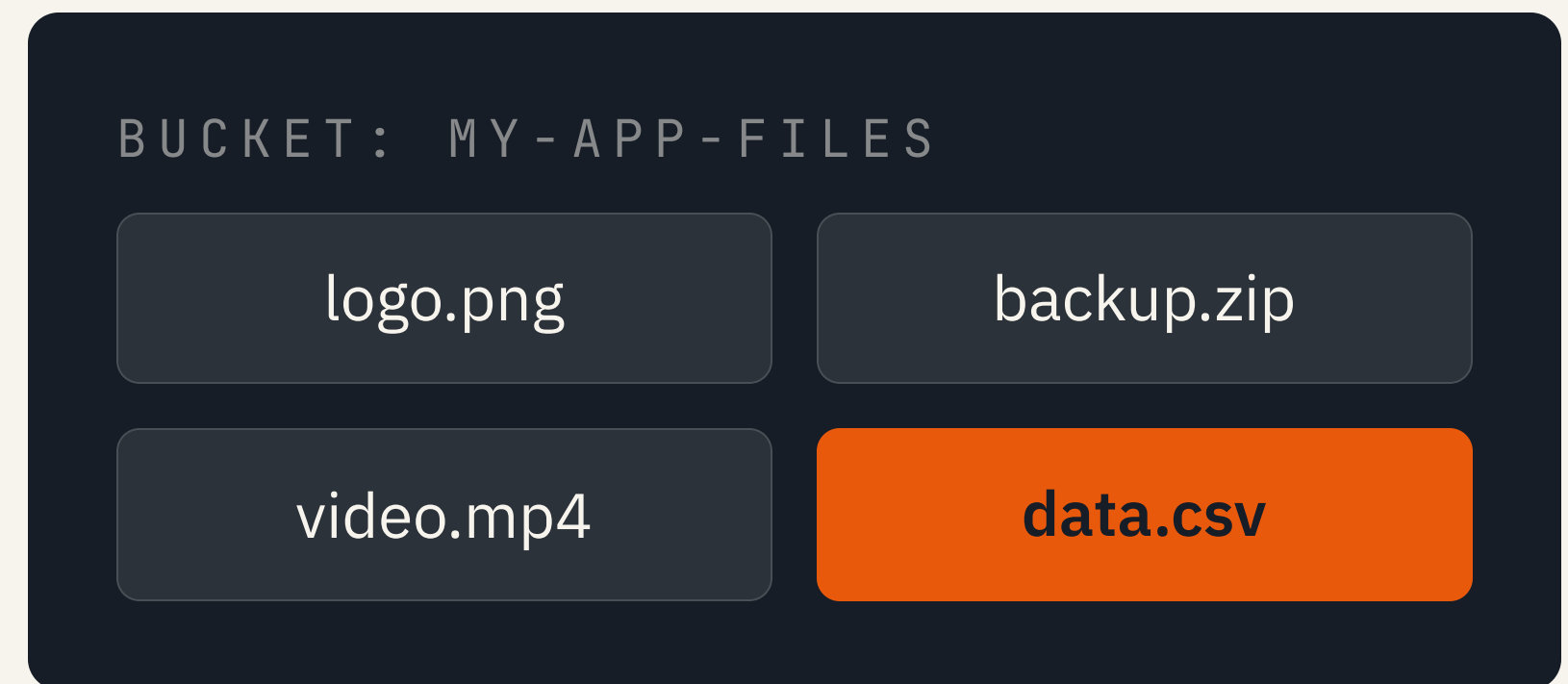
Operating System

Virtual Hardware (vCPU, RAM)

Cloud Storage

Cloud Storage holds files as objects inside buckets — images, backups, videos, anything.

- Scales to any amount of data
- Accessed over the internet by URL
- Storage classes for hot or cold data

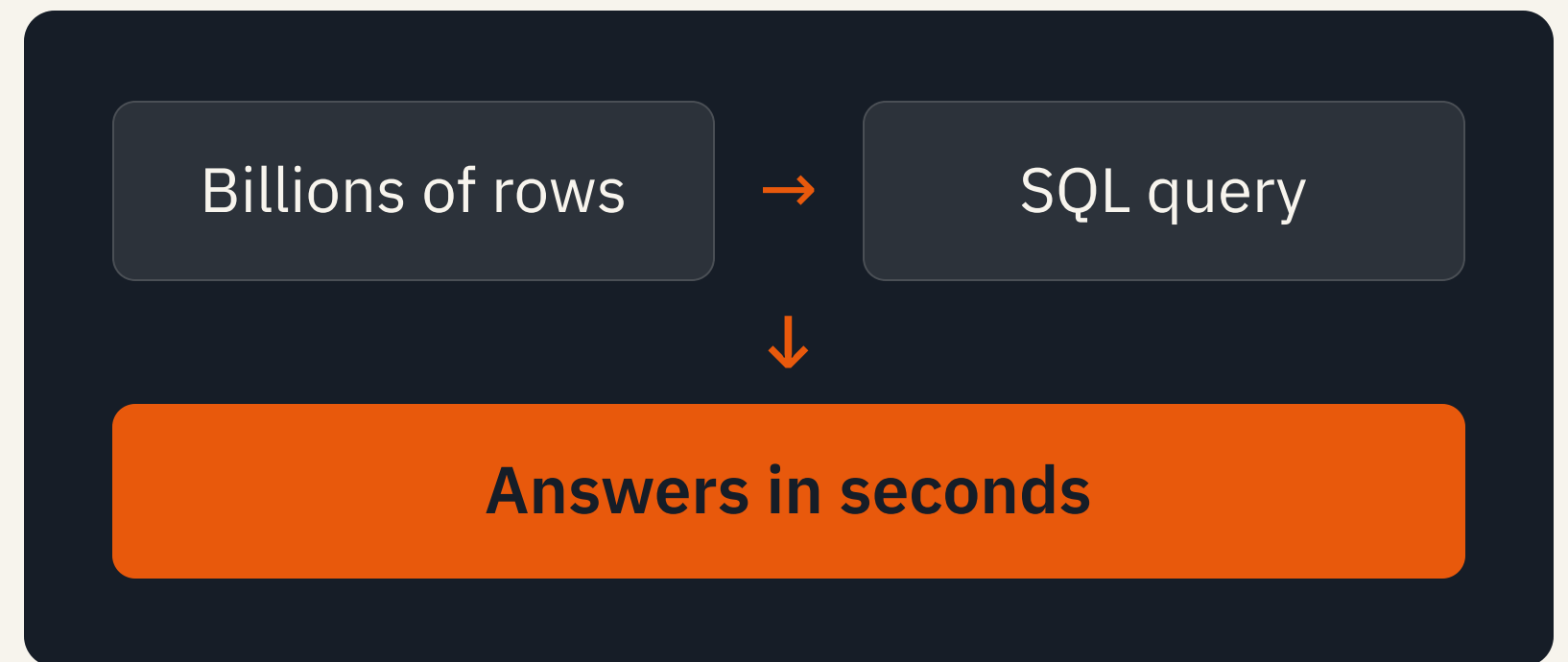


CORE SERVICE • ANALYTICS

BigQuery

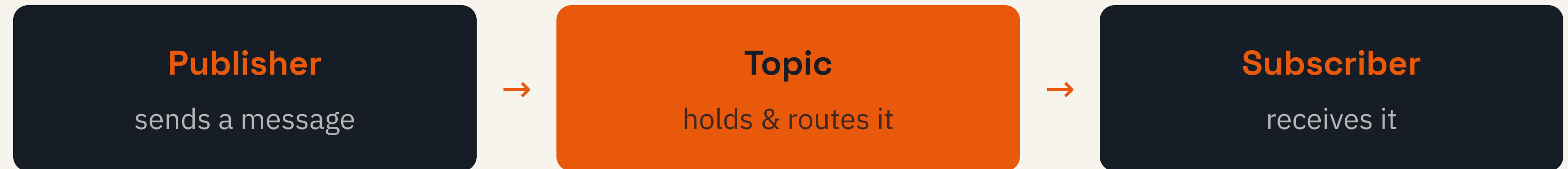
BigQuery is a data warehouse that runs SQL queries over huge datasets in seconds.

You load data, ask questions in SQL, and Google handles all the heavy lifting behind it.



Pub/Sub

Pub/Sub passes messages between services so they can talk without waiting on each other.

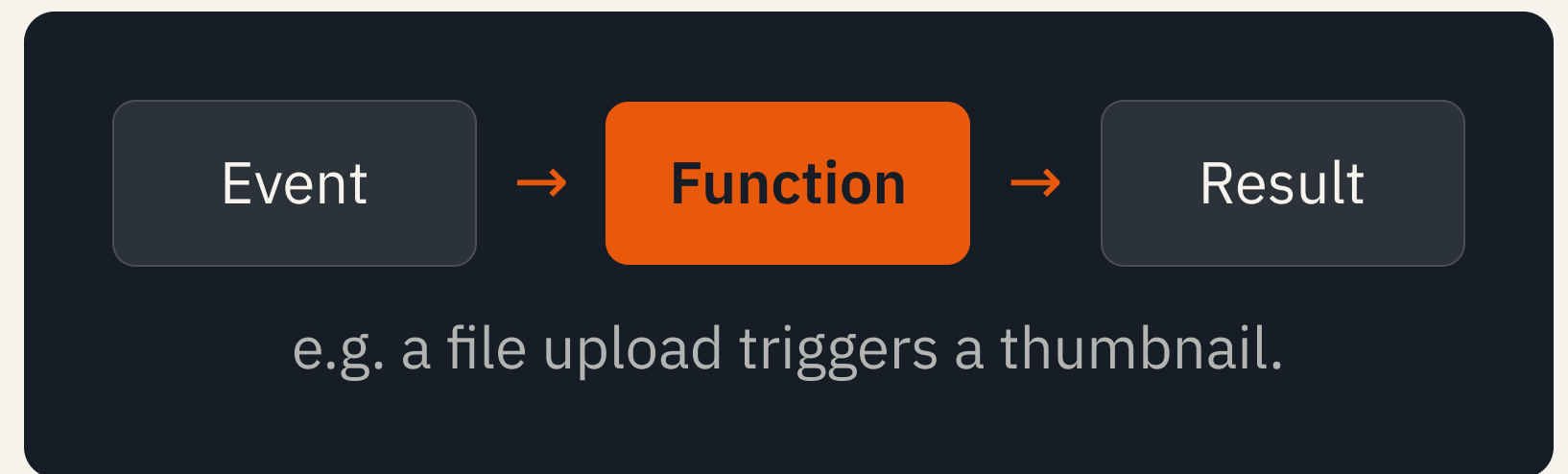


CORE SERVICE • SERVERLESS

Cloud Functions

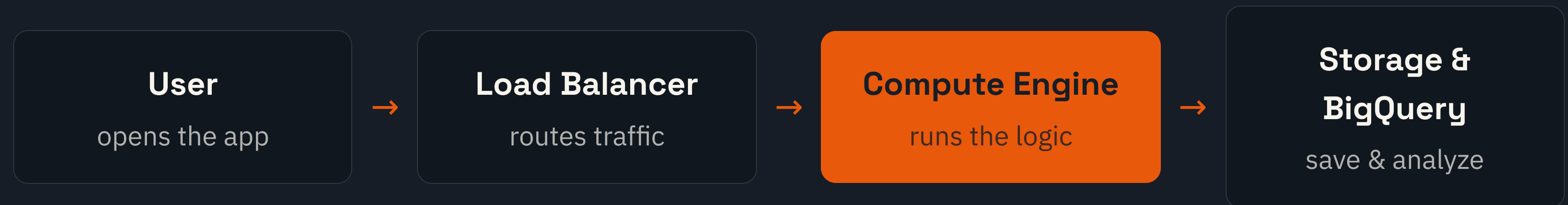
Cloud Functions run a small piece of code in response to an event — no servers to manage.

You write a function, GCP runs it when triggered, and you pay only while it runs.



PUTTING IT TOGETHER

How services work together



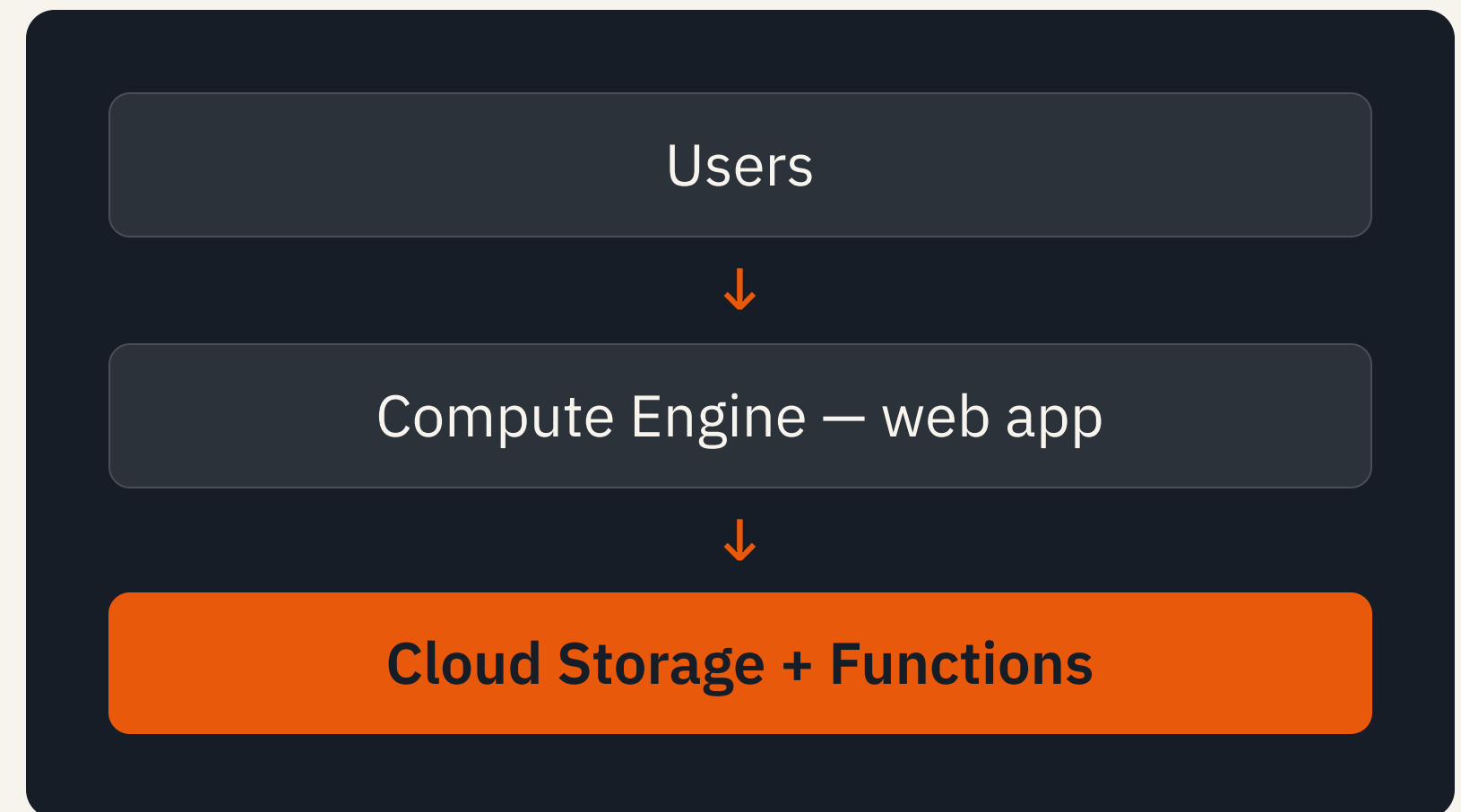
Many small services combine into one working application.

A REAL EXAMPLE

Example startup architecture

A small photo-sharing app might wire just a few GCP services together.

- Compute Engine serves the website
- Cloud Storage holds the photos
- Functions make thumbnails on upload



STAYING ONLINE

High availability

Redundancy

Run copies across zones so one failure does not stop the app.

Load balancing

Spread traffic evenly across healthy machines.

Auto scaling

Add machines when busy, remove them when quiet.

Design for failure – assume any single part can go down.

STAYING SAFE

Security basics

Least privilege

Grant only the access each identity truly needs.

Encryption

Data is encrypted at rest and in transit by default.

Audit logs

Every action is recorded so you can review it later.

Shared responsibility

Google secures the cloud; you secure what you build in it.

SPENDING WISELY

Cost awareness

Turn it off

Stop machines you are not using — idle resources still cost money.

Right-size

Pick a machine that fits the workload, not the biggest one.

Set budgets

Create alerts so you know before a bill grows too large.

The cloud is rented — every running resource is on the meter.

TIE IT TOGETHER

Revision mind map

Google Cloud Platform



Infrastructure

regions, zones, network

Organization

projects, billing, labels

IAM

members, roles, policy

Services

compute, storage, data

CHECK YOURSELF

Quick quiz

Q1

What is the difference between a region and a zone?

Q2

What does a project contain, and why does it matter?

Q3

In IAM, what links a member to a resource?

Q4

When would you use a service account instead of a user?

Q5

Name one service for compute, storage, and analytics.

Q6

Why do we run apps across multiple zones?

WHAT YOU LEARNED

Summary

Infrastructure

GCP runs on regions and zones across a global network.

Organization

Projects group resources and track billing and access.

IAM

Members get roles on resources — grant least privilege.

Services

Compute, storage, and analytics combine into real apps.

END OF MODULE

Questions & Discussion

You now know how GCP is built, organized, secured, and used.

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